AAIA Practice 1 - Work Report
Detailed Fuzzy Inference and Defuzzification

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1 Implemented Work and Pipeline

The implemented system detects three maneuvers from simulator telemetry sampled at 20 Hz:

- Stopping
- Overtaking
- U-Turnings

The final solution follows the execution chain below:

1. load telemetry file,
2. sort by elapsed time,
3. build sliding temporal windows,
4. extract window-level features,
5. compute fuzzy memberships,
6. perform rule aggregation and defuzzification,
7. convert fuzzy output to maneuver decision,
8. aggregate window decisions to timeline,
9. compare predictions with `Maneuver marker flag`.

Main implementation files:

- `stopping.py`: orchestration, windowing, voting, plotting, global configuration.
- `fuzzy_rules.py`: feature extraction, membership functions, fuzzy antecedents and scoring.

Generated outputs:

- `stopping_megagraph.png`
- `overtaking_megagraph.png`
- `u_turnings_megagraph.png`
- `metrics_results.txt`

2 Data Preparation and Temporal Segmentation

Data source structure used in the workspace:

- `data/Driver1` to `data/Driver5`
- One Excel file per maneuver and driver

Relevant files:

- `STISIMData_Stopping.xlsx`
- `STISIMData_Overtaking.xlsx`
- `STISIMData_U-Turnings.xlsx`

Main columns used by the final system:

- `Elapsed time`
- `speed`
- `Brake pedal`
- `Gas pedal`
- `Steering wheel angle`
- `Maneuver marker flag`

Preprocessing and segmentation steps applied:

1. Read each Excel file into a data frame.
2. Sort all rows by `Elapsed time`.
3. Generate fixed-length windows in seconds.
4. Discard windows with less than `min_samples`.
5. Keep overlap configurable through

$$\text{step_seconds} = \text{window_seconds} \cdot (1 - \text{overlap_ratio}).$$

3 Feature Engineering Per Window

For each temporal window, the system computes:

- mean acceleration: `mean_acc`,
- initial and final speed: `start_speed`, `end_speed`,
- speed dynamics: `speed_drop`, `speed_rise`,
- mean speed: `mean_speed`,
- braking behavior: `mean_brake`, `brake_rise`,
- throttle behavior: `mean_gas`,
- steering behavior: `mean_steering`, `end_steering`, `max_steering`,
- angular magnitude: `mean_abs_steering`,
- steering excursion: `steering_range`.

These features convert raw signals into interpretable linguistic evidence for fuzzy inference.

4 Detailed Fuzzy System Design

4.1 Membership functions

The implementation uses explicit membership mappings to convert crisp feature values into degrees in $[0, 1]$.

1) Positive saturation membership

$$\mu_+(x;t) = \begin{cases} 0 & x \leq 0 \\ \min\left(\frac{x}{t}, 1\right) & x > 0 \end{cases}$$

Used for evidence that should increase with larger positive values (for example acceleration, speed rise, brake intensity).

2) Negative saturation membership

$$\mu_-(x;t) = \begin{cases} 0 & x \geq 0 \\ \min\left(\frac{|x|}{t}, 1\right) & x < 0 \end{cases}$$

Used for deceleration evidence (negative acceleration).

3) Linear-below membership

$$\mu_{\text{below}}(x;m) = \max\left(0, \min\left(\frac{m-x}{m}, 1\right)\right)$$

Used to represent linguistic concepts such as *low speed* or *low gas*.

4) Mid-range triangular membership Given interval $[l, h]$, center $c = (l + h)/2$ and half-width $w = (h - l)/2$:

$$\mu_{\text{mid}}(x; l, h) = \max\left(0, 1 - \frac{|x - c|}{w}\right)$$

Used for values expected to be moderate (for example steering activity in overtaking).

4.2 Rule base: Stopping

Stopping combines positive evidence and one penalty:

- decelerating,
- reaches near zero speed,
- high brake,
- brake rise,
- intermediate transition,
- low gas,
- penalty if the window already starts near zero speed.

Detailed interpretation of stopping antecedents:

- **Decelerating** captures whether vehicle kinetics are consistent with a braking phase and not with cruising.
- **Reaches near zero speed** captures the temporal trend of a complete stop: start with nonzero speed and approach near-zero speed at the end of the window.
- **High brake** captures sustained brake engagement; this is one of the most discriminative cues in this maneuver.
- **Brake rise** captures the transition from little braking to strong braking, useful for detecting stopping onset.
- **Intermediate transition** captures realistic stopping behavior where the vehicle passes through intermediate speeds while decelerating.
- **Low gas** rejects windows where throttle remains high, reducing confusion with overtaking-like acceleration phases.
- **Start-near-zero penalty** avoids false positives on already-stopped windows that should not be interpreted as active stopping events.

The stopping fuzzy rule base (expressed linguistically) is:

- **R1:** IF deceleration is high THEN stopping evidence is high.
- **R2:** IF speed starts nonzero AND ends near zero THEN stopping evidence is high.
- **R3:** IF mean brake is high THEN stopping evidence is high.
- **R4:** IF brake increase is high THEN stopping evidence is medium-high.
- **R5:** IF mean speed is intermediate AND deceleration is present AND brake rise is present THEN stopping evidence is medium.
- **R6:** IF gas is low THEN stopping evidence is medium.
- **R7 (penalty):** IF start speed is already near zero THEN stopping evidence must be reduced.

In the final implementation, each rule contributes an activation level and a consequent label (low, medium, or high confidence), then Mamdani max-min aggregation is applied over the output universe.

Let α_r be the firing strength of rule r and $\mu_{C_r}(y)$ the consequent fuzzy set for confidence variable $y \in [0, 1]$. The implied output of each rule is:

$$\mu_r^{\text{out}}(y) = \min\left(\alpha_r, \mu_{C_r}(y)\right).$$

Then all rules are aggregated by:

$$\mu_{\text{agg}}(y) = \max_r \mu_r^{\text{out}}(y).$$

The crisp stopping confidence is obtained by centroid defuzzification:

$$S_{\text{stop}} = \frac{\int_0^1 y\, \mu_{\text{agg}}(y)\, dy}{\int_0^1 \mu_{\text{agg}}(y)\, dy}.$$

The final decision is:

$$\hat{y}_{\text{stop}} = \mathbb{1}[S_{\text{stop}} \geq \theta_{\text{stop}}], \quad \theta_{\text{stop}} = 0.70.$$

4.3 Rule base: Overtaking

Overtaking uses these antecedents:

- acceleration evidence,
- speed rise,
- sufficiently high mean speed,
- high gas,
- low brake,
- steering activity in a compatible range.

Detailed interpretation of overtaking antecedents:

- **Acceleration evidence** captures active engine demand and positive longitudinal dynamics.
- **Speed rise** captures net progression over the window, filtering out short acceleration spikes with no effective gain.
- **High mean speed** enforces contextual plausibility: overtaking is less likely at very low speeds.
- **High gas** strengthens confidence that the driver is intentionally increasing propulsion.
- **Low brake** rejects contradictory control patterns where braking dominates.
- **Steering activity in a compatible range** models that overtaking generally involves non-trivial lateral adjustment, but not the extreme steering profile of a U-turn.

The overtaking fuzzy rule base is:

- **R1:** IF acceleration is high THEN overtaking evidence is high.
- **R2:** IF speed rise is high THEN overtaking evidence is high.
- **R3:** IF mean speed is high THEN overtaking evidence is medium.
- **R4:** IF gas is high THEN overtaking evidence is medium-high.
- **R5:** IF brake is low THEN overtaking evidence is medium.
- **R6:** IF steering activity is in the overtaking-compatible range THEN overtaking evidence is medium.
- **R7 (penalty):** IF brake is high THEN overtaking evidence must decrease (low-confidence consequent).

Because overtaking can be confused with other transient behaviors (for example short acceleration bursts), the final system uses Mamdani max-min aggregation and centroid defuzzification instead of direct additive scoring. This reduces linear over-saturation and emphasizes coherent multi-rule support.

Steering activity is modeled as a weighted combination:

$$\text{steering_activity} = 0.6 \cdot \text{mean_abs_steering} + 0.4 \cdot \text{steering_range}.$$

Overtaking confidence is computed with the same Mamdani + centroid framework:

$$S_{\text{over}} = \frac{\int_0^1 y\, \mu_{\text{agg,over}}(y)\, dy}{\int_0^1 \mu_{\text{agg,over}}(y)\, dy}$$

with binary decision:

$$\hat{y}_{\text{over}} = \mathbb{1}[S_{\text{over}} \geq \theta_{\text{over}}], \quad \theta_{\text{over}} = 0.67.$$

4.4 Rule base: U-Turnings

U-turning evidence is dominated by steering geometry:

- high mean absolute steering,
- high steering range.

Detailed interpretation of U-turning antecedents:

- **High mean absolute steering** captures sustained steering commitment over the full window.
- **High steering range** captures large angular sweep, which is characteristic of U-turn execution.

The U-turning fuzzy rule base is:

- **R1:** IF mean absolute steering is high THEN U-turn evidence is high.
- **R2:** IF steering range is high THEN U-turn evidence is high.
- **R3 (implicit reinforcement):** IF both antecedents are high in the same window THEN combined weighted score exceeds threshold with high confidence.

This subsystem is intentionally compact and interpretable: it favors transparent geometric criteria over a large set of weaker secondary cues.

U-turning confidence is obtained with Mamdani aggregation and centroid defuzzification:

$$S_u = \frac{\int_0^1 y\, \mu_{\text{agg,u}}(y)\, dy}{\int_0^1 \mu_{\text{agg,u}}(y)\, dy}$$

and the decision is:

$$\hat{y}_u = \mathbb{1}[S_u \geq \theta_u], \quad \theta_u = 0.79.$$

5 Inference and Defuzzification in This Practice

The final project implementation uses Mamdani fuzzy inference with explicit centroid defuzzification over an output confidence universe in $[0, 1]$.

5.1 Fuzzification

Each crisp feature extracted from a window is converted to one or more membership degrees. Example: if a window has moderate positive acceleration, then $\mu_{\text{accelerating}}$ may be around 0.5 while other memberships have different values.

5.2 Rule evaluation and aggregation

Each rule computes a firing strength from antecedent memberships and rule weights. Then:

- implication uses min between firing strength and consequent set,
- aggregation uses max across all implied consequents.

Output consequents are represented by low/medium/high fuzzy sets (trapezoidal-triangular-trapezoidal).

5.3 Defuzzification

Defuzzification is performed through the centroid of the aggregated output membership:

- compute aggregated output fuzzy set on the confidence universe,
- compute centroid to obtain scalar confidence $S \in [0, 1]$,
- apply a threshold to obtain a binary decision.

Therefore, defuzzification in this implementation is the centroid-based mapping of aggregated output fuzzy sets into a crisp maneuver confidence score.

5.4 Temporal defuzzification at timeline level

A second crisping stage is applied after window-level inference:

$$\text{vote_ratio}(t) = \frac{\text{positive windows covering } t}{\text{total windows covering } t}$$

Then:

$$\hat{y}(t) = \mathbb{1}[\text{vote_ratio}(t) \geq \tau].$$

This resolves overlap conflicts and converts multiple window outputs into a sample-wise final decision.

6 Windowing Strategy and Overlap Analysis

Default configuration:

- Window length: 1.0 s
- Overlap ratio: 0.75 (step = 0.25 s)

Additional configurations used for mandatory analysis:

- Non-overlapping windows: overlap ratio 0.0
- Smaller windows: 0.5 s, overlap 0.75
- Larger windows: 1.5 s, overlap 0.75

7 Experimental Protocol and Metrics

Evaluation is computed against `Maneuver marker flag` with:

- Accuracy
- Sensitivity / TPR
- Specificity
- FPR

The script `eval_metrics.py` was executed and the output was saved to `metrics_results.txt`.

8 Quantitative Results (Latest Saved Run)

Table 1: Average metrics across drivers for each maneuver and configuration (from `metrics_results.txt`)

Maneuver	Configuration	Accuracy	TPR	Specificity	FPR
Stopping	1.0s, overlap 0.75	0.6879	0.5552	0.7366	0.2634
Stopping	1.0s, overlap 0.0	0.7248	0.4182	0.8368	0.1632
Stopping	0.5s, overlap 0.75	0.7024	0.4659	0.7889	0.2111
Stopping	1.5s, overlap 0.75	0.6921	0.6257	0.7167	0.2833
Overtaking	1.0s, overlap 0.75	0.6055	0.8422	0.4315	0.5685
Overtaking	1.0s, overlap 0.0	0.6055	0.8422	0.4315	0.5685
Overtaking	0.5s, overlap 0.75	0.6053	0.8308	0.4395	0.5605
Overtaking	1.5s, overlap 0.75	0.6152	0.8694	0.4272	0.5728
U-Turnings	1.0s, overlap 0.75	0.7572	0.6407	0.8602	0.1398
U-Turnings	1.0s, overlap 0.0	0.7542	0.5928	0.8882	0.1118
U-Turnings	0.5s, overlap 0.75	0.7469	0.6042	0.8671	0.1329
U-Turnings	1.5s, overlap 0.75	0.7682	0.6812	0.8506	0.1494

9 Interpretation Focused on Fuzzy Behavior

9.1 Stopping

Stopping is the most stable subsystem because its antecedents are highly coherent in real driving dynamics: deceleration, brake pressure, and low gas often appear together. This alignment strengthens fuzzy evidence and improves discrimination.

9.2 Overtaking

Overtaking is intrinsically harder because partial patterns appear in non-overtaking situations (brief acceleration, steering corrections, transient gas peaks). For this reason, overtaking required iterative retuning, explicit high-brake penalty logic, and centroid-based defuzzification to reduce oversaturation from simple additive accumulation.

9.3 U-Turnings

U-turning is mainly separable through steering intensity and steering excursion. Its fuzzy behavior is interpretable and robust; however, adding speed-shape evidence could reduce occasional false positives.

9.4 Overlapping vs non-overlapping windows

For Stopping and U-Turnings, overlap improves sensitivity and global accuracy. Non-overlapping windows increase specificity but miss more true maneuver samples.

For Overtaking, overlap comparison has almost no change because the active temporal logic already uses non-overlapping sequencing to enforce start-event behavior.

9.5 Effect of window size

Using 1.5 s windows improves TPR in all three maneuvers, indicating that larger temporal context helps fuzzy evidence accumulate.

Trade-off observed:

- larger windows: better sensitivity, sometimes lower specificity
- smaller windows: higher local specificity, lower sensitivity

9.6 Possible improvements

- Overtaking: add anti-false-positive fuzzy antecedents based on post-peak stabilization and return-to-lane pattern.
- U-Turnings: include speed-reduction and braking evidence jointly with steering to improve precision.
- Add explicit handling of gear transition rule (temporary gear=0 between two gears).
- Tune thresholds per driver or use validation split for robust global parameters.
- Tune output fuzzy set shapes (low/medium/high supports) per maneuver to improve calibration of centroid scores.

10 Visual Outputs

The following plots were generated and can be included in the final PDF:

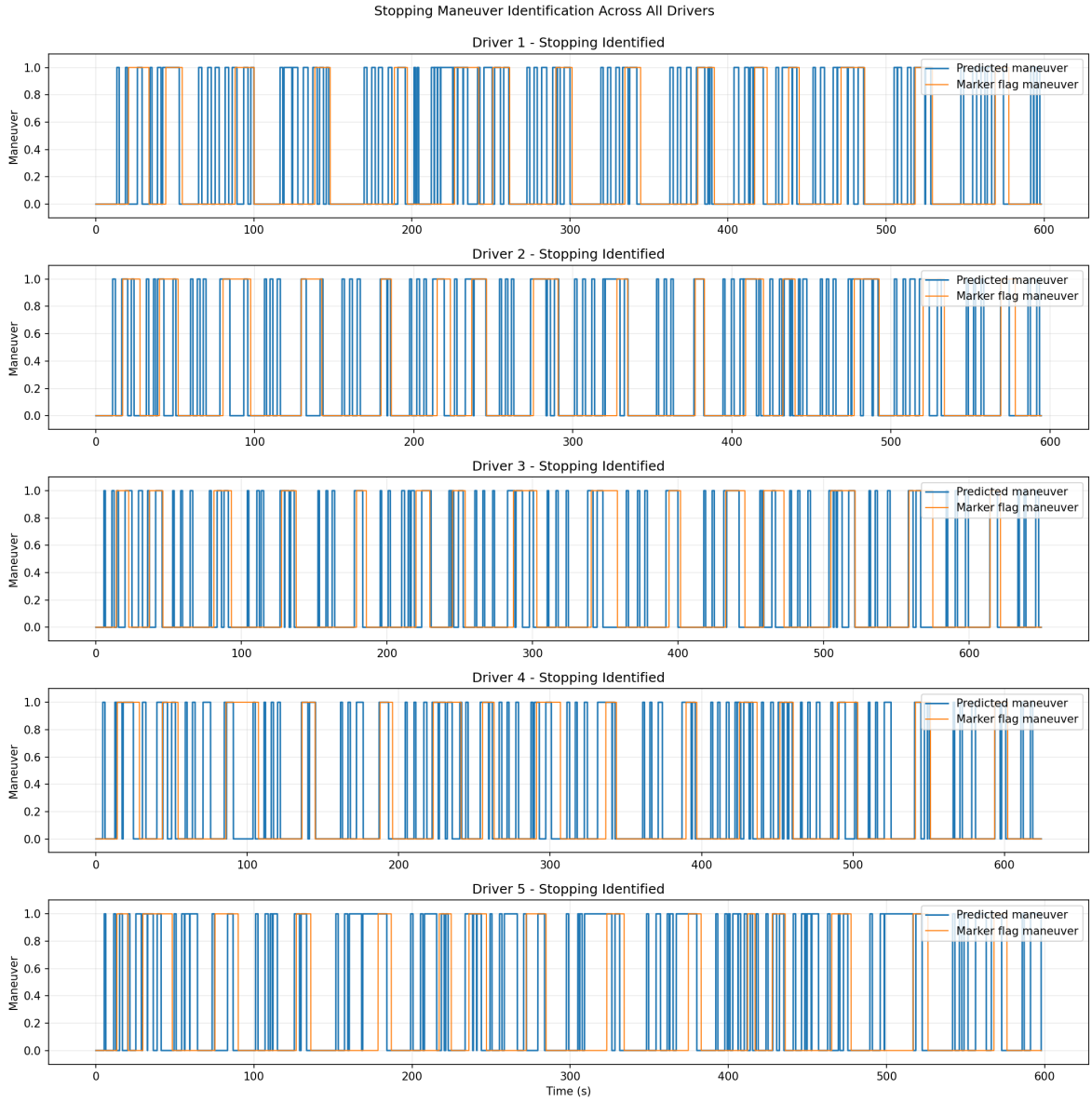

Figure 1: Stopping predictions vs marker flag across drivers

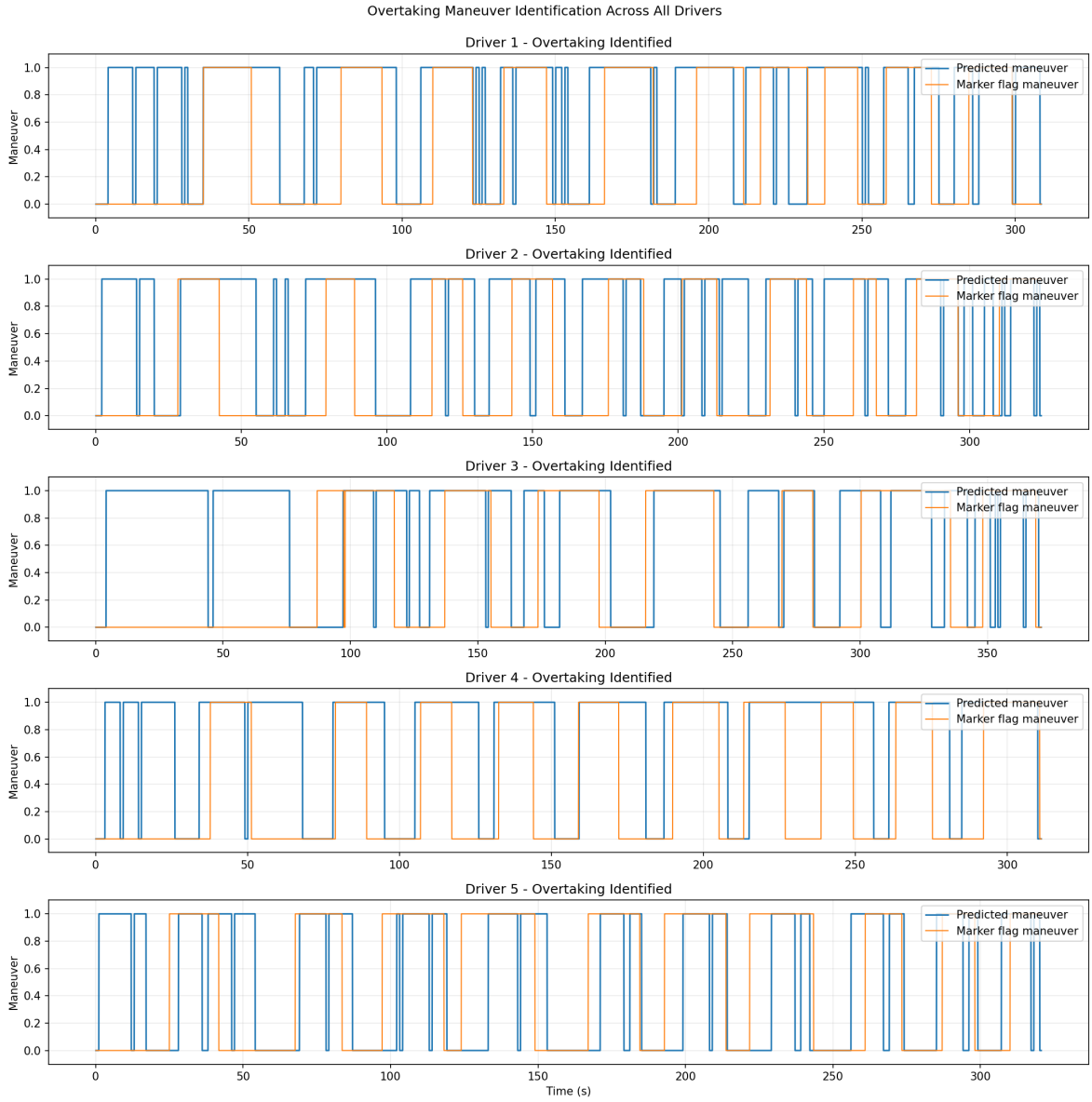

Figure 2: Overtaking predictions vs marker flag across drivers

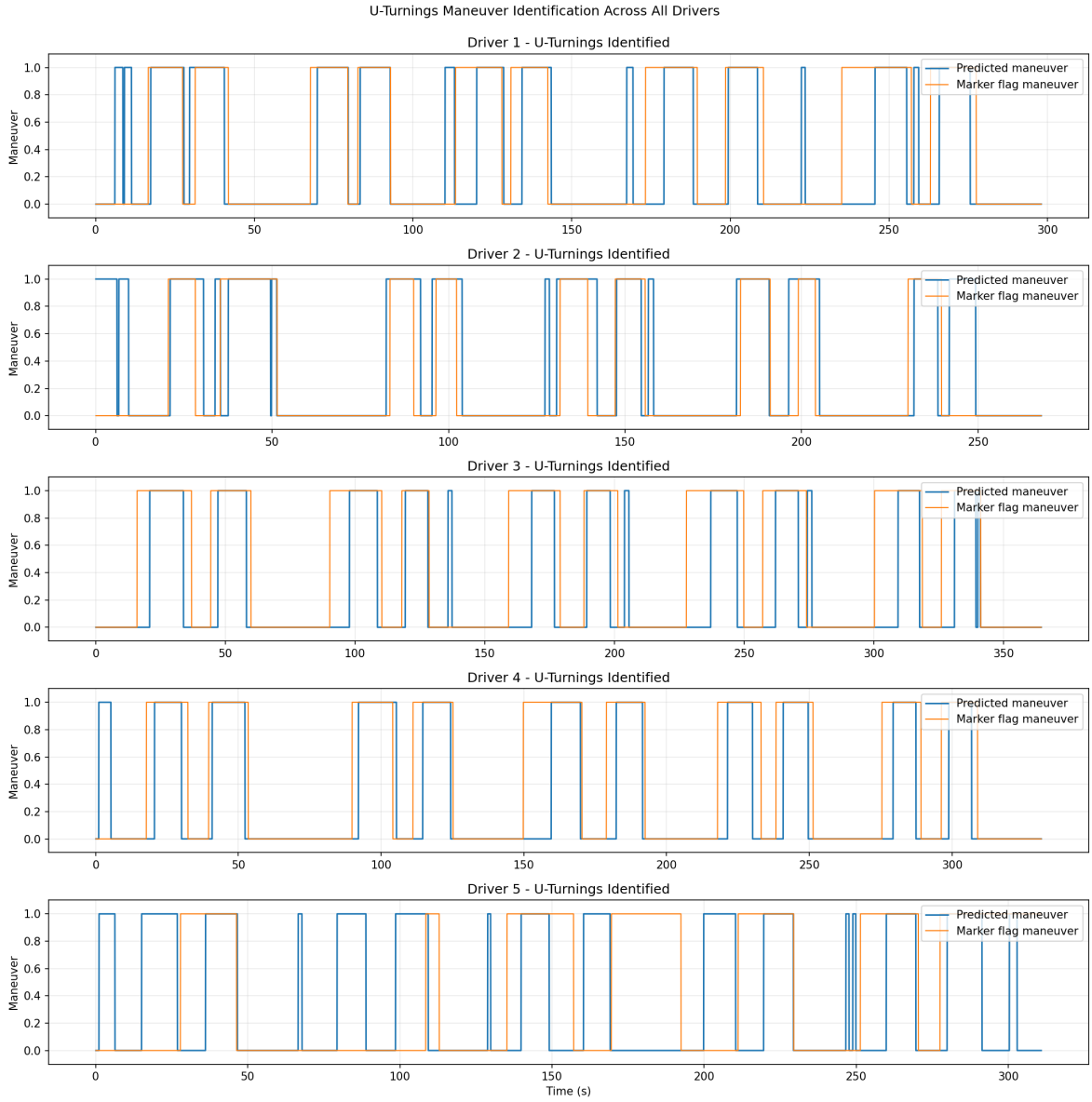

Figure 3: U-Turnings predictions vs marker flag across drivers

11 Reproducibility

Commands used to regenerate the final artifacts:

- Run full maneuver plots: `./.venv/bin/python stopping.py`
- Run evaluation metrics: `./.venv/bin/python eval_metrics.py > metrics_results.txt`